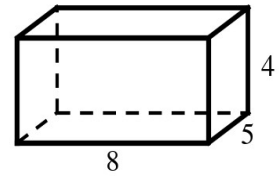


- A cube has 6 faces and 8 vertices. How many edges?
- A triangular prism has 5 faces and 6 vertices. Check Euler's formula.
- A pentagonal prism has 7 faces and 10 vertices. How many edges?

Example 2 - Cross Sections

- a) A plane slices a cylinder parallel to its bases. What is the cross section?
- b) A plane slices a cone through its vertex, perpendicular to the base.
- c) Any plane slices a sphere.



Example 3 - Rectangular Prism

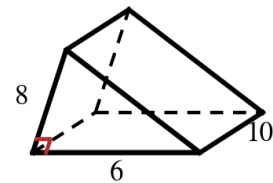
The prism at the right is 8 by 5 by 4.

- a) Find its volume.

Example 4 - Triangular Prism

The base is a right triangle with legs 6 and 8; the prism is 10 units long.

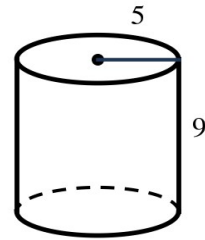
- a) Find the base area B , then the volume.



Example 5 - Cylinder

The cylinder at the right has radius 5 and height 9.

- a) Find its volume exactly, then to the nearest tenth.



Example 6 - Oblique Cylinder

The slanted cylinder at the right has radius 4 and height 7.

- a) Find its volume, and say why the slant doesn't matter.

Example 7 - Working Backward

- a) A prism with a 4-by-6 base has volume 96. Find its height.
- b) A cylinder with radius 6 has volume 252π . Find its height.
- c) A lap pool is 12 m by 8 m and 1.5 m deep everywhere.
How much water fills it?

